Project Management System

Software Requirements Specification

Team Details

Team No.	11
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Project Title	Project Management System
Description	Project Workflow, Visibility, Estimation, Kanban etc.
Tech Stack	Python (Django) / JavaScript (MERN Stack) or Any other Stack

1. Introduction

The Project Management System is a web-based application that helps teams plan, track, and manage their projects. It combines project workflow management, task visibility, effort estimation, and Kanban-style boards into one platform. Team members can create tasks, move them through workflow stages, estimate effort, and collaborate through comments and notifications, while project managers get a clear view of overall project progress.

2. Functional Requirements

ID	**Requirement**	**Description**
FR1	**User Authentication**	Users can register, log in, log out, and reset passwords. Role-based access for Admin, Project Manager, and Team Member.
FR2	**Project Management**	Users can create, edit, archive, and delete projects, and add or remove team members.
FR3	**Task/Issue Management**	Create, edit, delete, and assign tasks with title, description, priority, due date, and labels.
FR4	**Kanban Board**	Visualize tasks in columns (Backlog, To Do, In Progress, Review, Done) with drag-and-drop status updates.
FR5	**Workflow Management**	Define custom workflow states and valid transitions between them for each project.
FR6	**Estimation**	Assign story points or time estimates to tasks and track estimated versus actual effort.

ID	Requirement	Description
FR7	**Sprint/Milestone Management**	Create sprints or milestones, add tasks to a sprint backlog, and close sprints.
FR8	**Visibility & Dashboard**	Show project progress, burndown/velocity charts, and task distribution by status or assignee.
FR9	**Notifications**	Notify users on task assignment, status changes, comments, and approaching deadlines.
FR10	**Comments & Collaboration**	Users can comment on tasks and attach files or documents.
FR11	**Search & Filter**	Search tasks or projects and filter by assignee, priority, status, label, or sprint.
FR12	**Reporting**	Generate downloadable reports such as progress summaries, velocity, and workload.
FR13	**Activity Log**	Maintain an audit trail of changes made to a task or project.

3. Non-Functional Requirements

ID	Category	Requirement
NFR1	**Performance**	Page loads and Kanban updates should respond within 2 seconds under normal load.
NFR2	**Scalability**	The system should support a growing number of projects and users without major redesign.
NFR3	**Security**	Passwords must be hashed, access must be role-based, and the system must use HTTPS and guard against SQL injection, XSS, and CSRF.
NFR4	**Usability**	The interface should be intuitive, support drag-and-drop, and be responsive on desktop and tablet.
NFR5	**Reliability**	The system should maintain at least 99% uptime with graceful error handling.
NFR6	**Maintainability**	The codebase should be modular (MVC/MVT pattern), with documented APIs and unit test coverage.
NFR7	**Portability**	The system should work across major browsers and on Windows, Linux, and macOS.
NFR8	**Data Backup & Recovery**	The system should perform regular automated backups and support recovery in case of failure.
NFR9	**Interoperability**	The system should expose a REST API for integration with tools such as Slack, GitHub, or email.